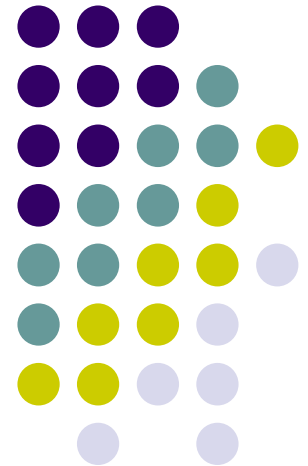
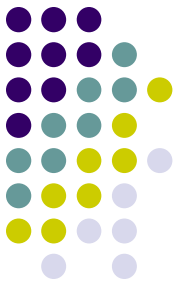


Chapter 10: Hierarchical Generalized Linear Models



Conceptual and Statistical Background



- Consider a simple level-1 model

$$Y_{ij} = \beta_{0j} + \beta_{1j}(SES)_{ij} + r_{ij}$$

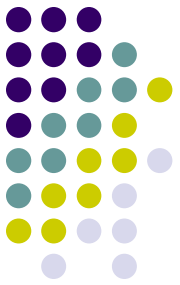
where

$$Y_{ij} = \begin{cases} 1 & \text{if retained in grade} \\ 0 & \text{if not} \end{cases}$$

$$E[Y_{ij} | \beta_{0j}, \beta_{1j}] = \beta_{0j} + \beta_{1j}(SES)_{ij} = \phi_{ij}$$

where

ϕ_{ij} = the person – specific predicted probability of retention



Problems with this setup

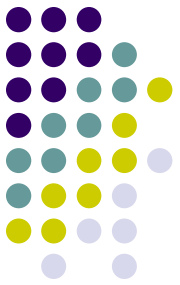
- HLM assumes the level-1 random effect r_{ij} to be normal with constant variance. But,

- Level-1 random effect has two discrete values:

- $$r_{ij} = Y_{ij} - [\beta_{0j} + \beta_{1j}(SES)_{ij}] = \begin{cases} 1 - [\beta_{0j} + \beta_{1j}(SES)_{ij}] \\ -[\beta_{0j} + \beta_{1j}(SES)_{ij}] \end{cases}$$

- Level-1 variance is heterogeneous:

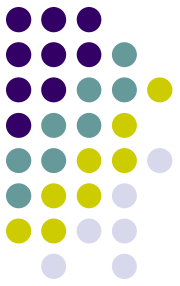
- $$Var(Y_{ij} | \beta_{0j}, \beta_{1j}) = \phi_{ij}(1 - \phi_{ij})$$



Moreover....

- The Linear Model for ϕ_{ij} actually makes no sense:
 - Each unit increase in SES should lead to a smaller change in ϕ_{ij} as ϕ_{ij} approaches 0 or 1.
- Therefore, we move to a “generalized linear model” rather than a standard linear model at level 1

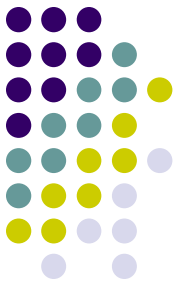
The Three Parts of the Level-1 Model in HGLM



- **Sampling model** – describes the distribution of the level-1 observations
- **Link function** – transforms the model-based predicted values to be within the required range
- **Structural model** – same as we have seen in conventional HLM models

(Level 2 model remains the same as in standard HLM-multivariate normal random effects)

HLM as a Special Case of HGLM



- Level-1 Sampling Model:

$$Y_{ij} \mid \mu_{ij} \sim NID(\mu_{ij}, \sigma^2)$$

$$E(Y_{ij} \mid \mu_{ij}) = \mu_{ij}, \quad \text{Var}(Y_{ij} \mid \mu_{ij}) = \sigma^2$$

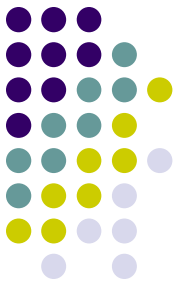
- Level-1 Link Function:

$$\eta_{ij} = \mu_{ij}$$

- Level-1 Structural Model:

$$\eta_{ij} = \beta_{0j} + \beta_{1j} (SES)_{ij}$$

Revised level-1 model for binary outcome



- Level-1 Sampling Model:

$$Y_{ij} \mid \phi_{ij} \sim \text{Bernoulli}(\phi_{ij})$$

$$E(Y_{ij} \mid \phi_{ij}) = \phi_{ij}, \quad \text{Var}(Y_{ij} \mid \phi_{ij}) = \phi_{ij}(1 - \phi_{ij})$$

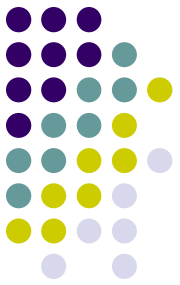
- Level-1 Link Function:

$$\eta_{ij} = \log\left(\frac{\phi_{ij}}{1 - \phi_{ij}}\right)$$

- Level-1 Structural Model:

$$\eta_{ij} = \beta_{0j} + \beta_{1j}(\text{SES})_{ij}$$

HGLM for binary outcome (cont)



- Level-1 Link Model

$$\exp(\eta_{ij}) = Odds = \frac{prob}{1 - prob} = \frac{\phi_{ij}}{1 - \phi_{ij}}$$

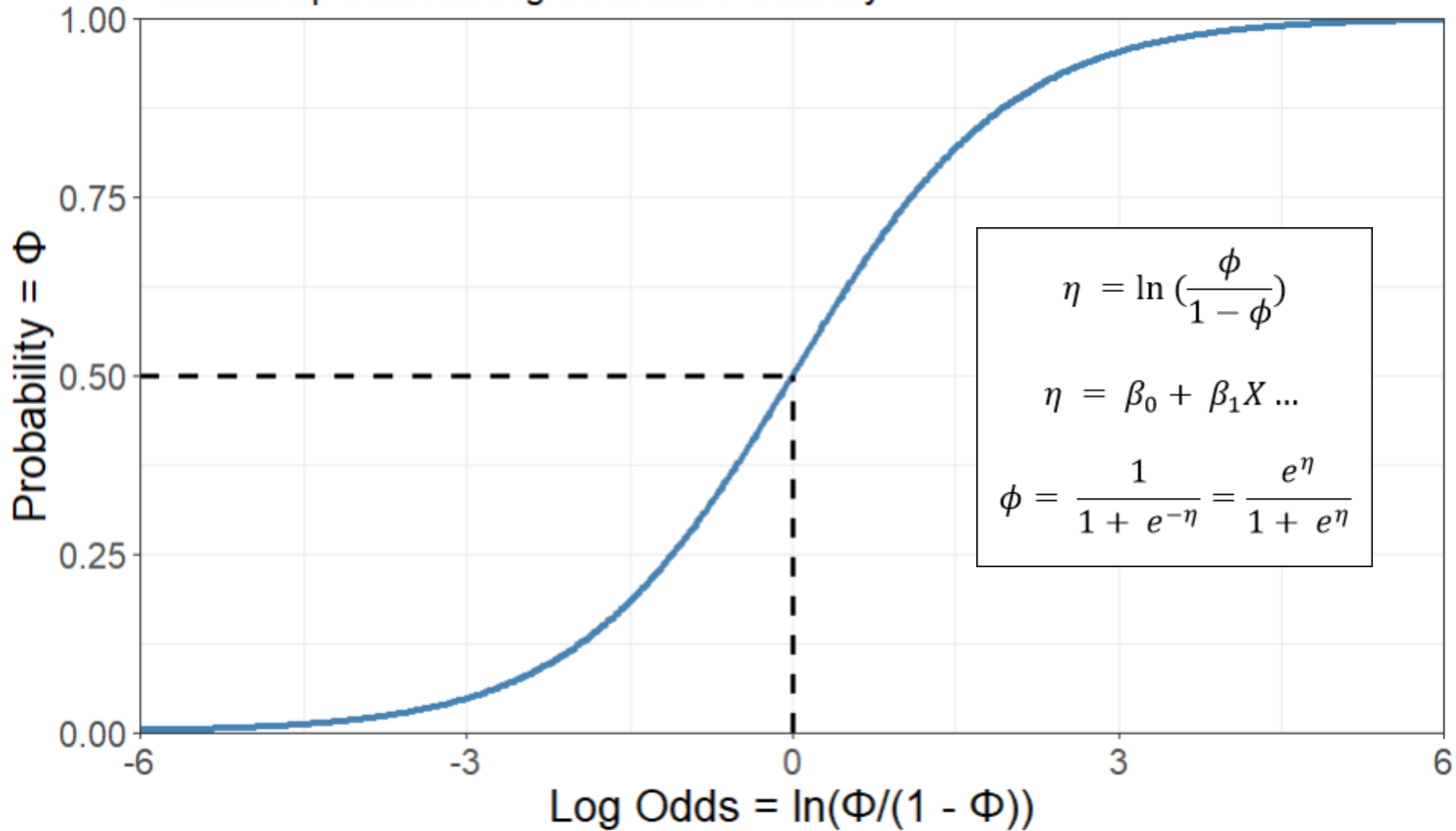
$$\eta_{ij} = \log\left(\frac{\phi_{ij}}{1 - \phi_{ij}}\right): \text{``log odds (logits = log odds units)''}$$

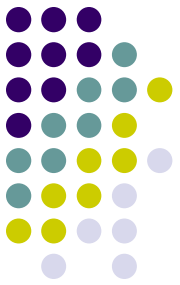
- In reverse

$$\text{probability from log odds: } \phi_{ij} = \frac{1}{1 + \exp(-\eta_{ij})} = \frac{\exp(\eta_{ij})}{1 + \exp(\eta_{ij})}$$



Relationship Between Log Odds and Probability

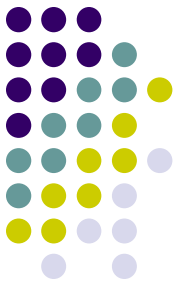




Conversion Table

Probability ϕ	Odds $\frac{\phi}{1 - \phi}$	Log odds $\ln(\frac{\phi}{1 - \phi})$
$< .5$	< 1	Negative
$.5$	1	0
$> .5$	> 1	Positive

Bernoulli Example: Grade Retention in Thailand

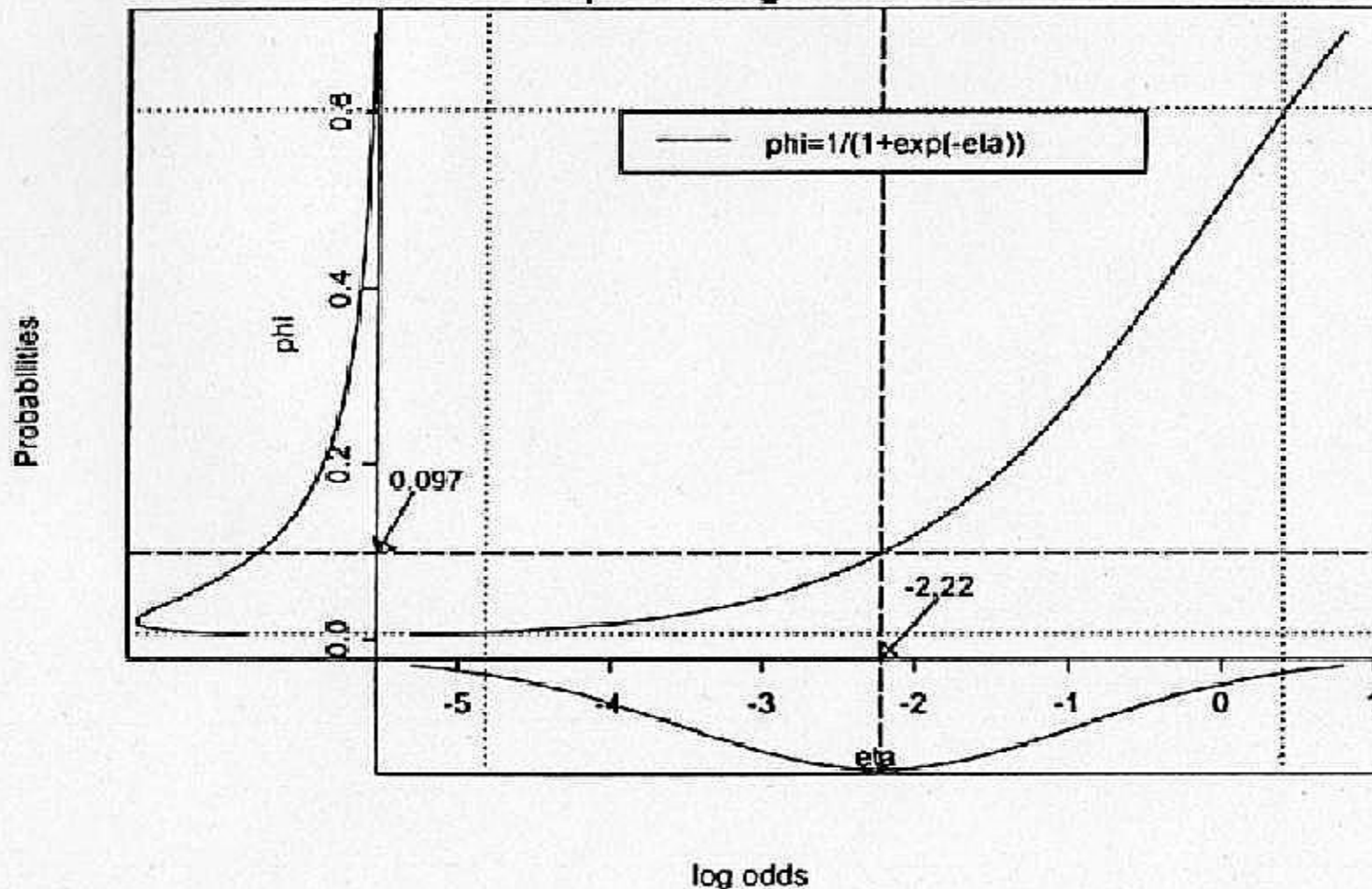


- Outcome: Grade repetition in primary grades (14%)
- Level-1 Predictors: Sex, pre-primary experience
- Level-2 Predictors: Mean SES

Probability and Log Odds of Retention

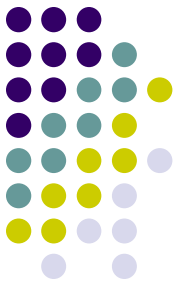


Figure 10.1 : School-specific probability of retention as a function of school-specific log-odds of retention



$$95\% p.v., \eta_{ij} = -2.22 \pm 1.96(1.70)^{.5} = (-4.78, 0.34)$$

$$95\% p.v., \phi_{ij} = (0.008, 0.59)$$



Unit-specific model

$$\eta_{ij} = \beta_{0j} = \gamma_{00} + u_{0j}$$

$$E(Y_{ij} | u_{0j}) = \frac{1}{1 + \exp[-(\gamma_{00} + u_{0j})]}$$

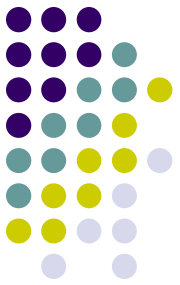
$$\hat{E}(Y_{ij} | u_{0j} = 0) = \frac{1}{1 + \exp(-\hat{\gamma}_{00})} = \frac{1}{1 + \exp(-2.22)} = .097$$

Population-average model

$$E(Y_{ij}) = \frac{1}{1 + \exp(-\gamma_{00}^*)}$$

$$\hat{E}(Y_{ij}) = \frac{1}{1 + \exp(-\hat{\gamma}_{00}^*)} = \frac{1}{1 + \exp(-1.72)} = .14$$

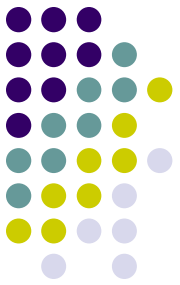
*These numbers are from the textbook. The data used in class are slightly different, so estimates won't match exactly.



Interpreting Probability Changes

When you present findings for a logistic regression, it is advisable to translate effects back into changes in probabilities, which most people can understand.

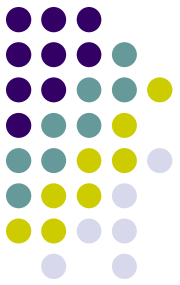
But remember, **the change in probabilities associated with any coefficient will be different depending on the values of all X values in the model.**



Interpreting Probability Changes

You therefore need to specify the conditions under which your probability-change calculations will be valid.

This is often best accomplished by centering to give an useful intercept.



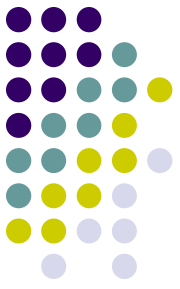
Interpreting Probability Changes

Be especially careful to distinguish between *percent* changes in the probability, and *percentage point* changes in the probability.

A probability increase from .2 to .3 represents:

- A 50 *percent* increase in probability
- A 10 *percentage-point* increase in probability

Which is more appropriate depends on context.

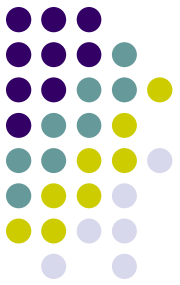


Odds Ratios

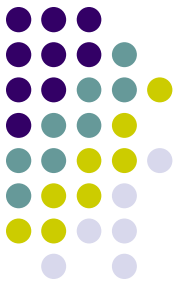
Sometimes we look at odds ratios instead of coefficients for logistic regressions. In some fields, they are more common than discussing changes in log odds/ logits.

$$\text{Odds Ratio} = \frac{\left(\frac{\phi_{Y|x=1}}{1 - \phi_{Y|x=1}} \right)}{\left(\frac{\phi_{Y|x=0}}{1 - \phi_{Y|x=0}} \right)}$$

Odds Ratios



- These describe the *multiplicative* change in odds for a one unit change in a predictor.
 - E.g. for a two-unit change in the predictor, the odds change by $OR \times OR$ (*not* $2 \times OR$, as with coefficients in log odds units)
- Odds ratios range from 0 to infinity.
- Odds ratios >1 indicate increased chances of Y occurring. Odds ratios <1 indicate decreased chances of Y occurring.
- The odds ratio for a $\beta = \exp(\beta) = e^\beta$

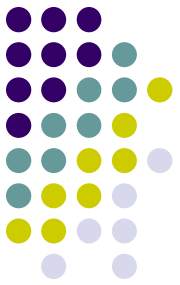


Odds Ratios

Advantage: The odds ratio for a one-unit change in a predictor will be the same regardless of your other predictors, as long as they are held constant.

Disadvantages: They are only really interpretable to those who are already familiar with odds ratios.

Odds ratios can be very large, but represent small changes in probability (in situations where probabilities are very low).



Count Data

- Consider a simple level-1 model

$$Y_{ij} = \beta_{0j} + \beta_{1j}(SES)_{ij} + r_{ij}$$

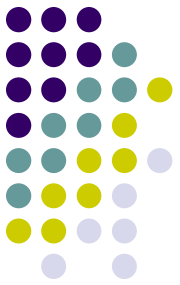
where

$Y_{ij} \in \{0, 1, 2, \dots\}$ number of days absent

$$E[Y_{ij} \mid \beta_{0j}, \beta_{1j}] = \beta_{0j} + \beta_{1j}(SES)_{ij} = \lambda_{ij}$$

where

λ_{ij} = the person - specific predicted event rate



Problems with this setup

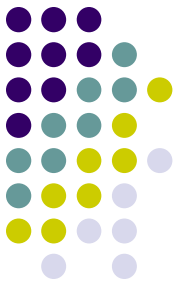
- HLM assumes the level-1 random effect r_{ij} to be normal with constant variance. But,

- Level-1 random effect will be discrete:
- Level-1 variance is heterogeneous:

$$\text{Var}(Y_{ij} \mid \beta_{0j}, \beta_{1j}) = \lambda_{ij}$$

- The Linear Model for λ_{ij} actually makes no sense:
 - Each unit increase in SES should lead to a smaller change in λ_{ij} as λ_{ij} approaches 0.

Revised level-1 model for count outcome



- Level-1 Sampling Model:

$$Y_{ij} \mid \lambda_{ij} \sim \text{Poisson}(\lambda_{ij})$$

$$E(Y_{ij} \mid \lambda_{ij}) = \lambda_{ij}, \quad \text{Var}(Y_{ij} \mid \lambda_{ij}) = \lambda_{ij}$$

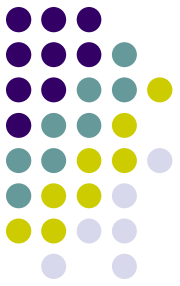
- Level-1 Link Function:

$$\eta_{ij} = \log(\lambda_{ij})$$

- Level-1 Structural Model:

$$\eta_{ij} = \beta_{0j} + \beta_{1j}(\text{SES})_{ij}$$

Level-1 model for count outcome (variable exposure)



- Level-1 Sampling Model:

$$Y_{ij} \mid \lambda_{ij} \sim \text{Poisson}(m_{ij}\lambda_{ij})$$

$$E(Y_{ij} \mid \lambda_{ij}) = m_{ij}\lambda_{ij}, \quad \text{Var}(Y_{ij} \mid \lambda_{ij}) = m_{ij}\lambda_{ij}$$

where m_{ij} is exposure

- Level-1 Link Function:

$$\eta_{ij} = \log(\lambda_{ij})$$

- Level-1 Structural Model:

$$\eta_{ij} = \beta_{0j} + \beta_{1j}(\text{SES})_{ij}$$